1. Experiment in short

Number of nodes	4
Number of topologies	6
Location of the experiment	Haifa, Israel $(32.471285°\text{N}, 34.876159°\text{E})$
Measurement duration	Between 30 and 90 minutes
Measurement granularity	5 seconds
Link quality measurements	Bit Error Ratio

2. Location

The experiment took place in May 2009 in the Haifa harbor, Israel. The harbor depth is 12 m. The harbor walls and piers are made of hard concrete, and acoustic link connection is available only in line-of-sight conditions. A Google map illustration of the experimental setup is shown in Fig. 1.



Fig. 1. Satellite picture of the sea trial location (picture taken from Google maps on September 29, 2009). The different locations of the vessels (and thus network nodes) are marked.

3. Topologies

The network included four nodes, deployed from 4-m rubber boats. The nodes changed their locations to form the topologies illustrated in Fig. 2. In each location, the boats anchored with a long rope that allowed location shifts of up to 20 m. The following table describes the boats' locations with respect to the positions marked in Fig. 1. Topology 1 is a fully connected network; Topology 2 is a partially connected network; Topology 3 is a star topology centered around node 2; Topology 4 is a tandem network; Topology 5 is a similar tandem network, but node 1 is located far from its one-hop neighbour node 2; finally, Topology 6 is a partial tandem network with node 1 disconnected from the other nodes.

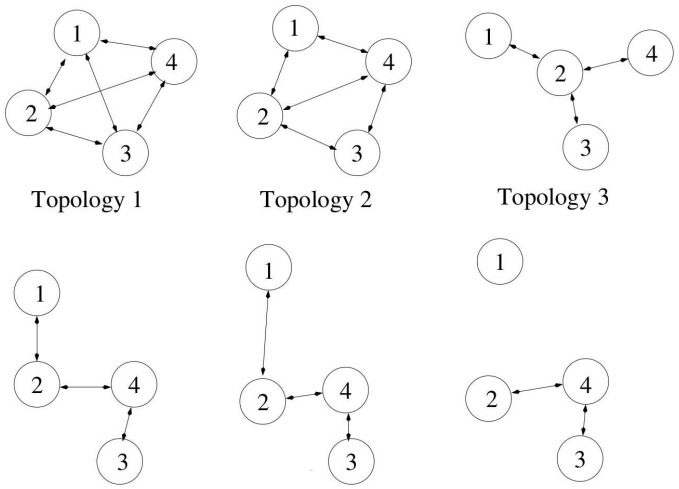


Fig. 2. Topologies tested during the sea trial corresponding to vessel locations shown in Fig. 1.

TABLE I

NODE LOCATIONS FOR THE SIX TOPOLOGIES WITH RESPECT TO THE MARKS IN FIG. 1

Topology	**Node 1**	**Node 2**	**Node 3**	**Node 4**	**Duration**
1 (all connected)	1A	2A	3A	4A	30 minutes
2 (partial connection)	1B	2A	3B	4A	60 minutes
3 (star)	1B	2A	3B	4B	60 minutes
4 (tandem)	1B	2A	3C	4B	60 minutes
5 (tandem with near-far)	1C	2A	3C	4B	60 minutes
6 (partial tandem)	1D	2A	3C	4B	90 minutes

4. DESCRIPTION OF THE NODES

Each node had an ITC transceiver, operating at the 20 kHz-30 kHz frequency band. The transceivers were deployed at depth of 4 m, and were connected through a T/R switch to a B&K power amplifier with an input and an output to a 100k samples per second D/A - A/D National Instrument data acquisition, which, in turn, was operated by a laptop computer on board each vessel. Transmission power was 165 dB Re 1 μPa@1m. GPS receivers kept track on the location of each boat.

The nodes transmitted a B-PSK signal modulated by a direct sequence spread spectrum signal, which

was created using a gold sequence-based pseudo rand sequence of 128 chips. The receiver employed a simple phase-locked-loop mechanism along with a decision feedback equalizer. For each packet, a training sequence of 10 symbols was perceived by a 20kHz-30kHz linear frequency modulation signal for time synchronization. No channel coding was used, and the effective transmission rate was 600 bps.

5. Tested protocols and schemes

The nodes operated a spatial reuse time-devision-multiple-access (TDMA) medium access control. Each node was given a designated time slot of 5 s to transmit a broadcast message. By evaluating the time-varying topology, nodes could share transmissions in the same time slot while avoiding collisions.

6. Link quality measurements

The network is evaluated by its links' bit error ratio (BER). Upon transmission, the BER for the link between the transmitter and the different receivers was evaluated directly from the de-modulation process. The result is a time-varying per-link BE for all six tested topologies.

7. Available data

The measured data is provided in file `HaifaHarborExp-05-09.mat`, which includes all topologies presented in Fig. 2. For each topology, we provide 951 measurement epochs, which have been taken at constant intervals of 1 second starting from the beginning of each experiment. The file contains four types of matrices:

- the location matrices `LocMat1`...`LocMat6`, of size 360×4×3 (`LocMat1`), 720×4×3 (`LocMat2` to `LocMat5`), and 1080×4×3 (`LocMat6`);
- the topology measurement matrices `TopMat1`...`TopMat6`, of size 360×4×4 (`TopMat1`), 720×4×4 (`TopMat2` to `TopMat5`), and 1080×4×4 (`TopMat6`);
- the adjacency matrices `AdjMat1`...`AdjMat6` of size 360×4×4 (`AdjMat1`), 720×4×4 (`AdjMat2` to `AdjMat5`), and 1080×4×4 (`AdjMat6`);
- the technology matrices `TechMat1`...`TechMat6` of size 360×4×1 (`TechMat1`), 720×4×1 (`TechMat2` to `TechMat5`), and 1080×4×1 (`TechMat6`);.

Additionally, for each matrix type, we concatenate all matrices into a single matrix, respectively named:

- `FullLocMat` of size 4320×4×3;
- `FullTopMat` of size 4320×4×4;
- `FullAdjMat` of size 4320×4×4;
- `FullTechMat` of size 4320×4×1.

Using Matlab/Octave matrix notation:

- `LocMat1(t,i,:)...LocMat6(t,i,:)` or `FullLocMat(t,i,:)` are the UTM coordinates (zone 36S) and the depth of node i at time instant t;
- `TopMat1(t,i,j)...TopMat6(t,i,j)` or `FullTopMat(t,i,j)` is the BER for link $i \rightarrow j$ at time t;
- `AdjMat1(t,i,j)...AdjMat6(t,i,j)` or `FullAdjMat(t,i,j)` is equal to 1 if there exists a topology link between nodes i and j at time t;
- `TechMat1(t,i,p)...TechMat6(t,i,p)` or `FullTech(t,i,p)` is equal to 1 if node i can use physical layer technology p at time t.

8. Reference to detailed description

Further details about this experiment are given in the following papers:

- R. Diamant, G. N. Shirazi, L. Lampe, "Robust Spatial Reuse Scheduling in Underwater Acoustic Communication Networks," *IEEE Journal of Oceanic Engineering*, vol. 39, no. 1, pp. 32–46, Jan. 2014.
- R. Diamant, L. Lampe, "Spatial Reuse Time-Division Multiple Access for Broadcast Ad Hoc Underwater Acoustic Communication Networks," *IEEE Journal of Oceanic Engineering*, vol. 36, no. 2, pp. 172–185, Apr. 2011.